



Natural Language Processing: Seq2seq and Attention

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Machine Learning and Data-Intensive Systems

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- **Sequence to Sequence (seq2seq)**
- LSTM
- Attention
- Practical tips

Machine translation requires training parameters to provide results

Human Translation

$$y^* = \arg \max_y p(y|x)$$

The “probability” is intuitive and is given by a human translator’s expertise

Machine Translation

model parameters

$$y' = \arg \max_y p(y|x, \theta)$$

Questions we need to answer

- **modeling**

How does the model for $p(y|x, \theta)$ look like?

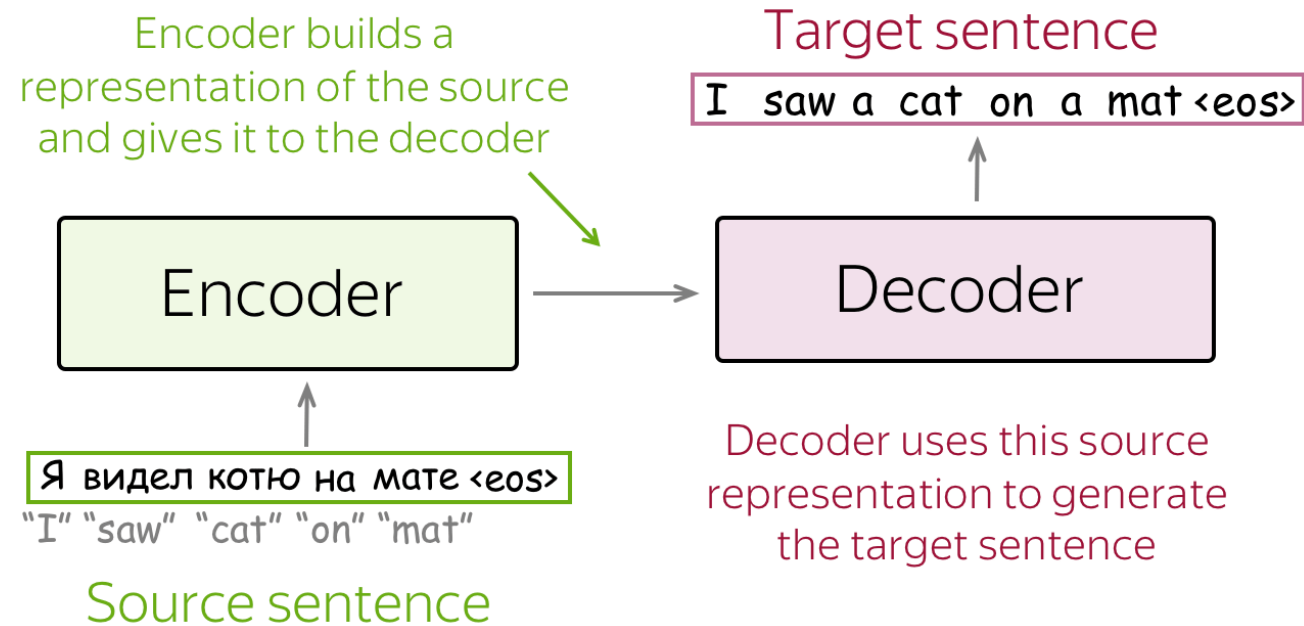
- **learning**

How to find θ ?

- **search**

How to find the argmax?

Encoder-decoder architecture maps data semantics



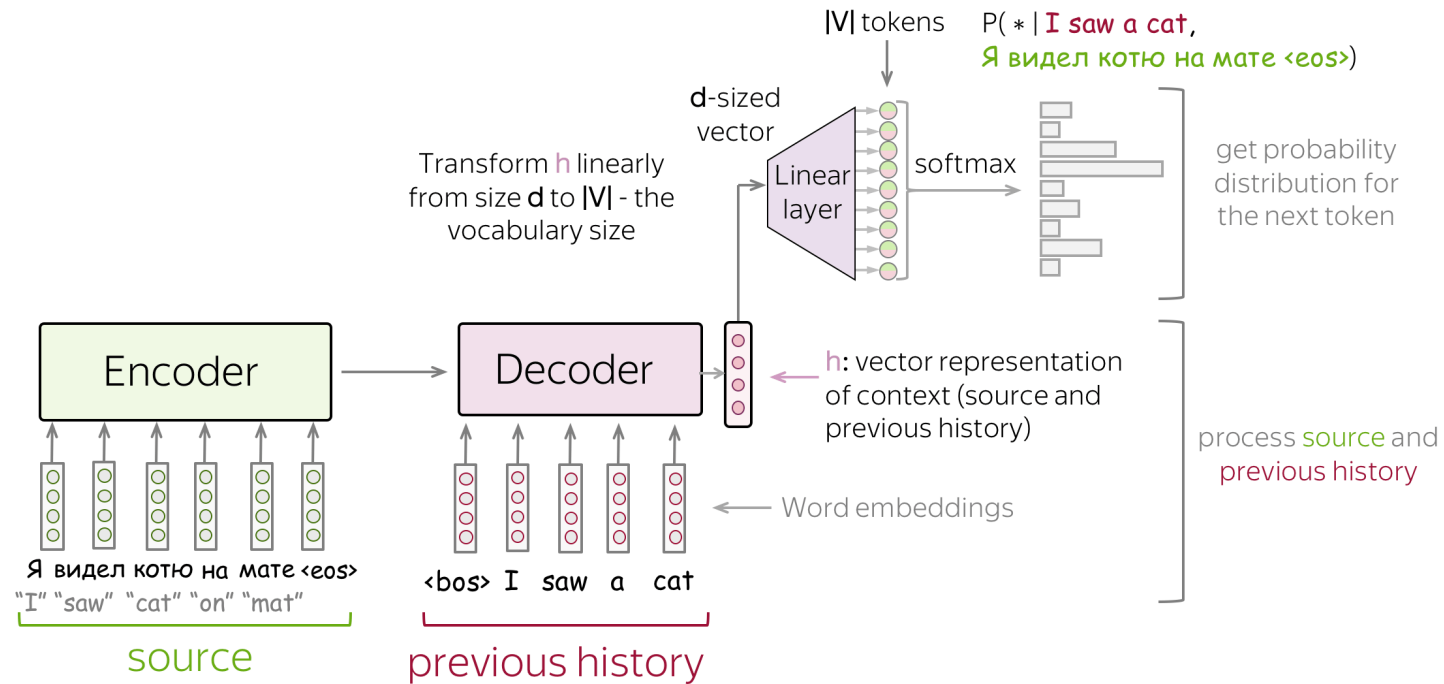
Encoder encapsulates a condition for a decoder Language Model

Language Models:
$$P(y_1, y_2, \dots, y_n) = \prod_{t=1}^n p(y_t | y_{<t})$$

Conditional
Language Models:
$$P(y_1, y_2, \dots, y_n, | \textcolor{green}{x}) = \prod_{t=1}^n p(y_t | y_{<t}, \textcolor{green}{x})$$


condition on source x

A helicopter view on the Encoder-Decoder architecture



Model Loss is a well-known Cross-Entropy

Source sequence:

Я видел котю на мате <eos>
"I" "saw" "cat" "on" "mat"

Target sequence:

I saw a cat on a mat <eos>

← one training example

← one step for this example

previous tokens we want the model to predict this

Model prediction: $p(* | \text{I saw a, Я ... <eos>})$

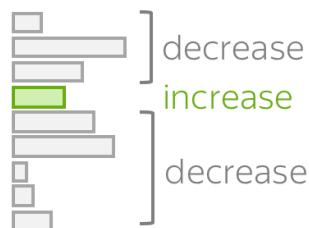


Target

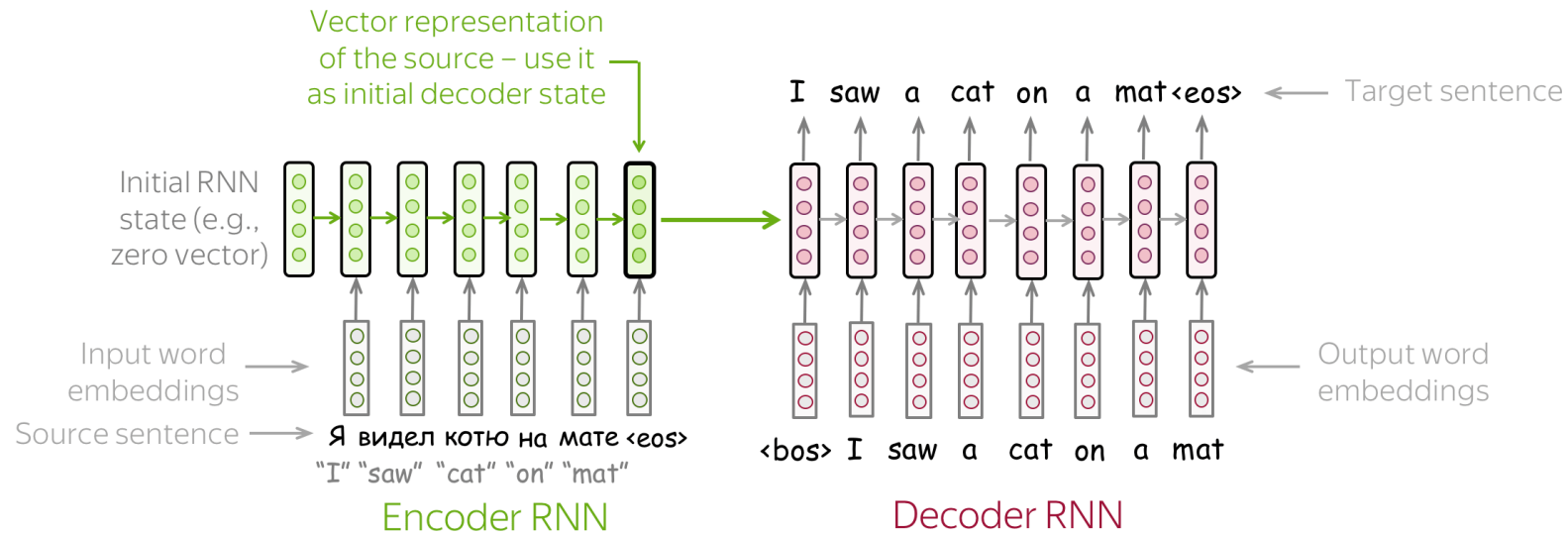
0
0
0
0
1
0
0
0
0
0
0

cat

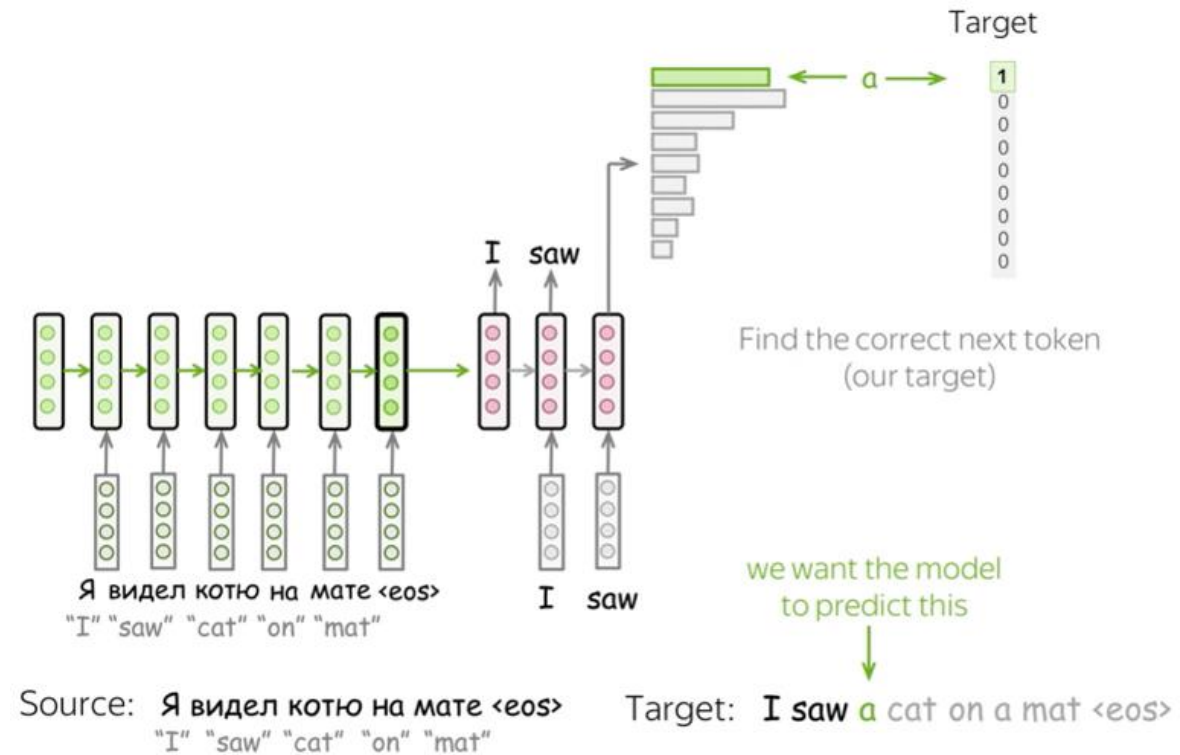
Loss = $-\log(p(\text{cat})) \rightarrow \min$



A simple RNN is a valid Encoder-Decoder model



A simple RNN is a valid Encoder-Decoder model



Semantic space of text embeddings

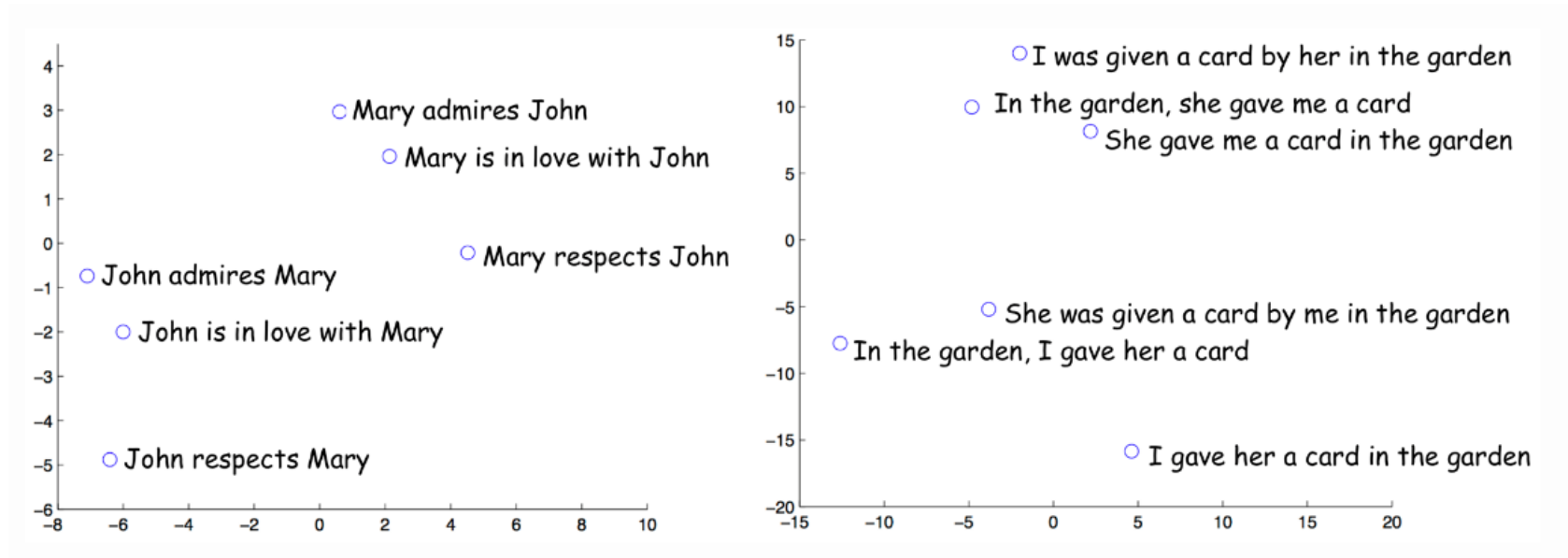
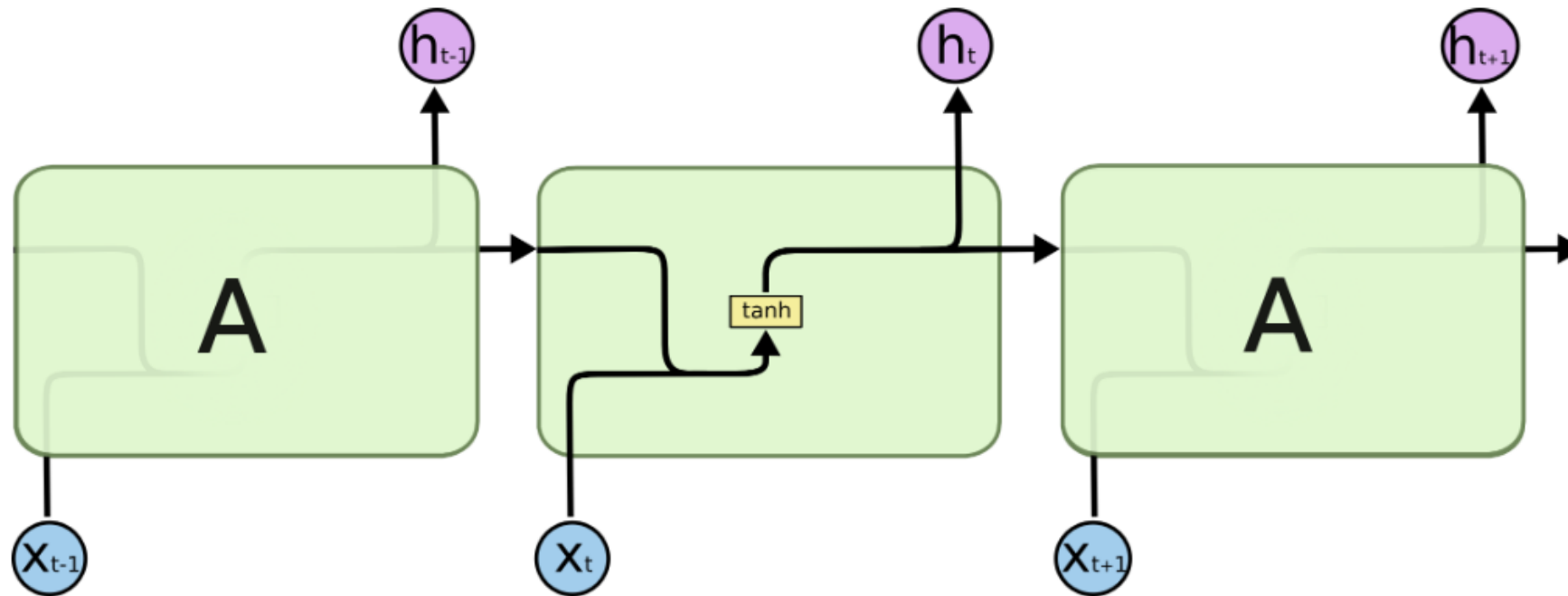




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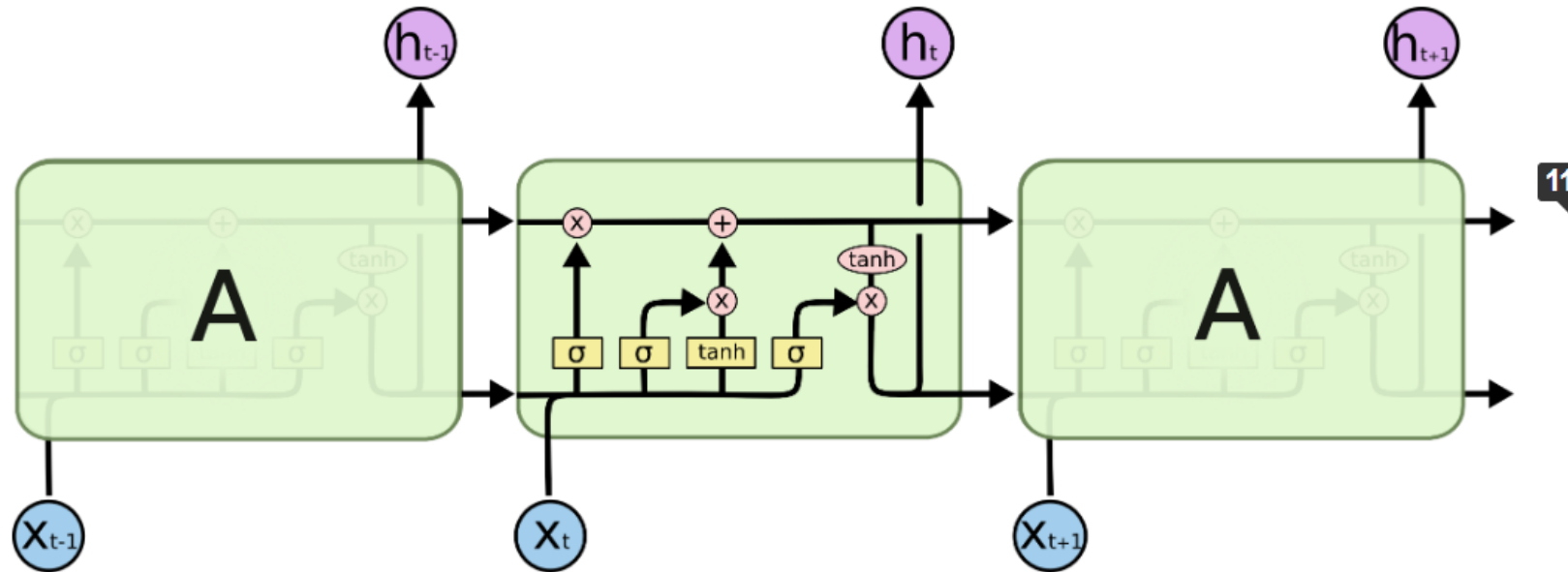
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A vanilla RNN fails to account for long-term information



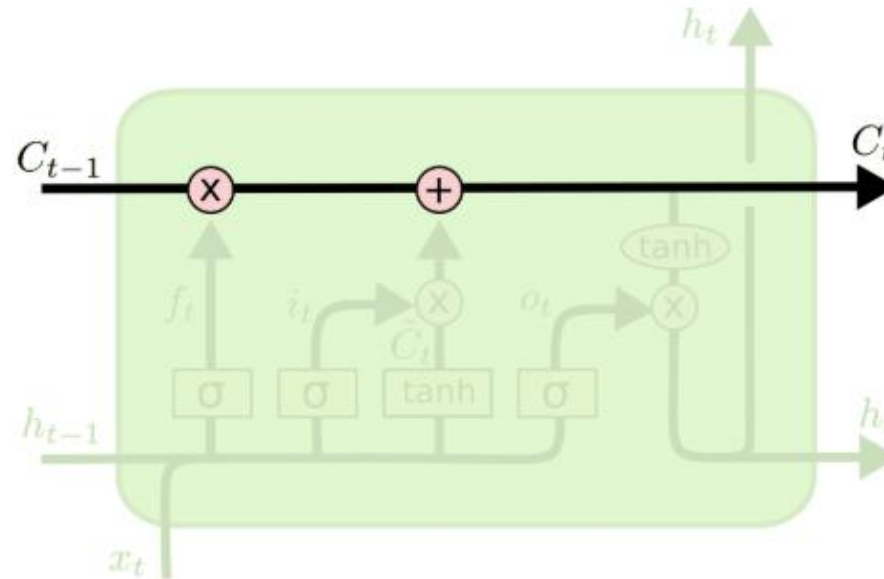
The repeating module in a standard RNN contains a single layer.

LSTM builds on an idea of RNN, introducing a separate channel for knowledge transmission

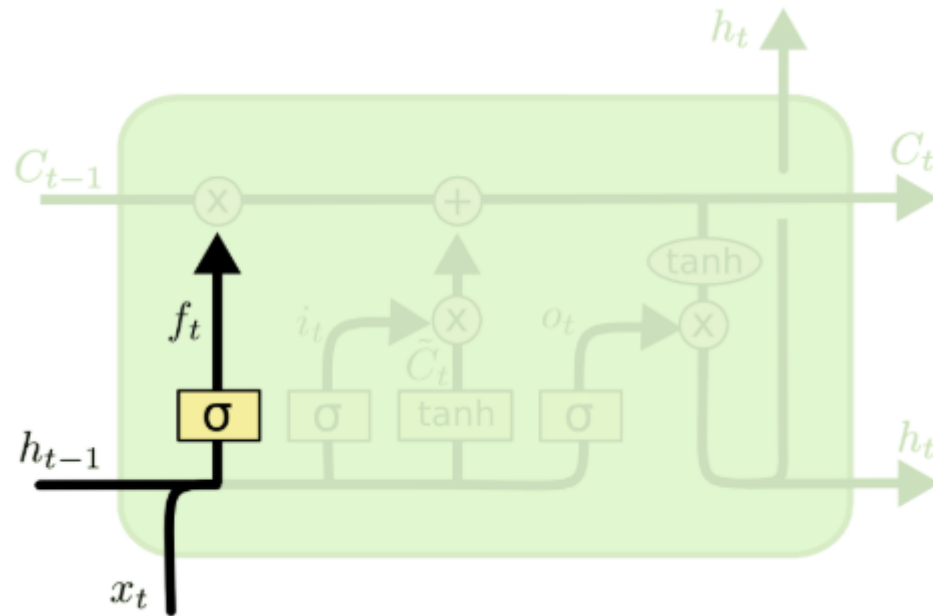


The repeating module in an LSTM contains four interacting layers.

The information flow is supported by the C channel

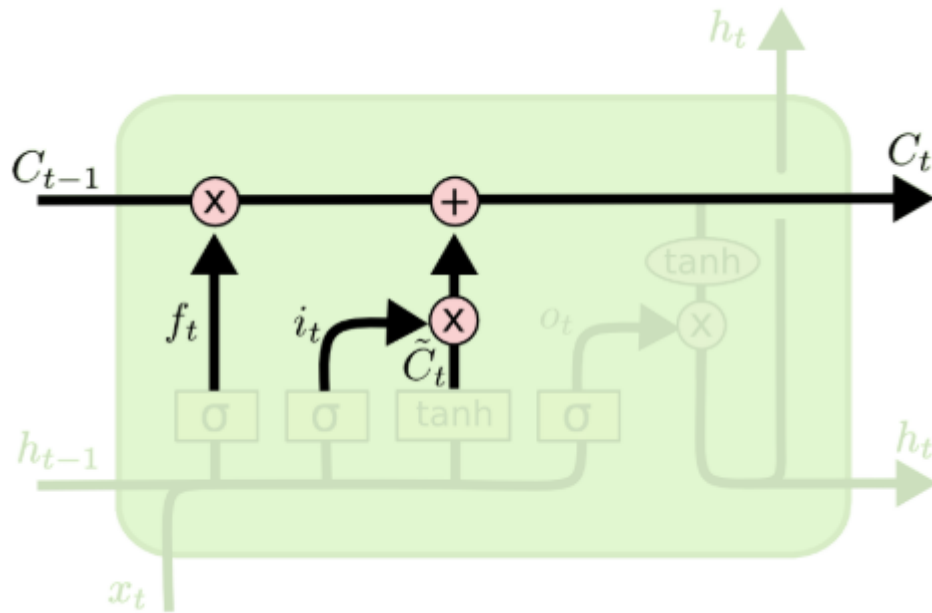


The idea is to control what new information is relevant to C



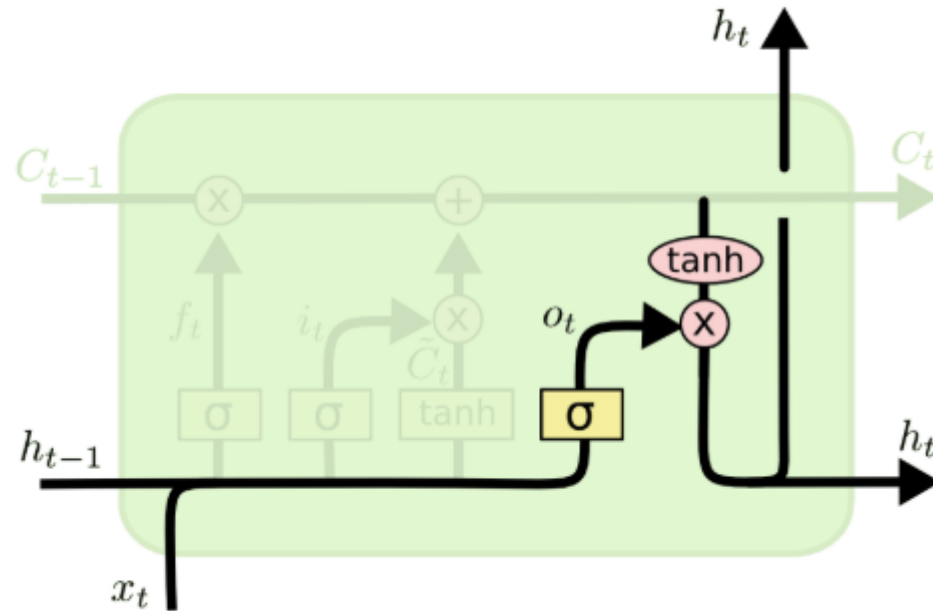
$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

The idea is to control what new information is relevant to C



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

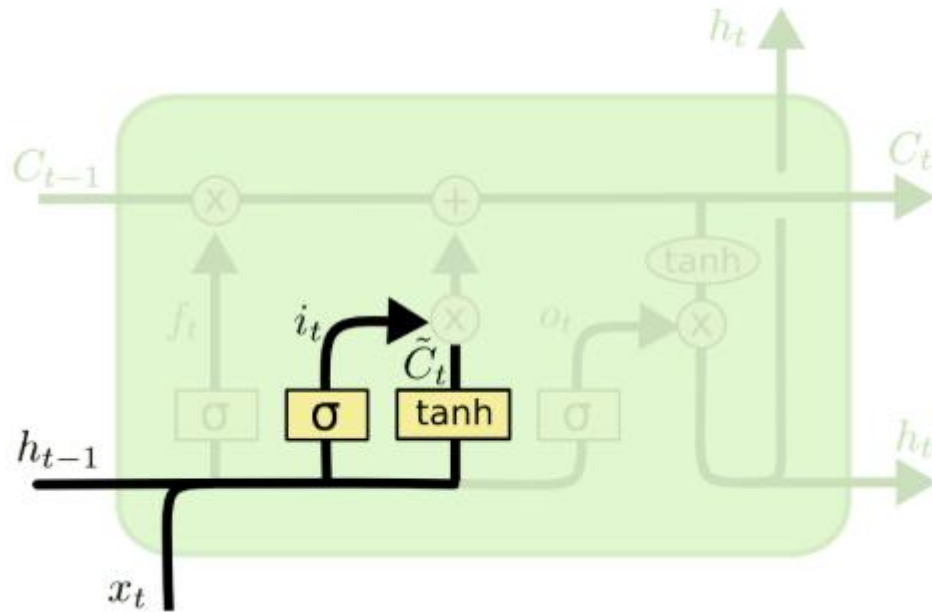
The idea is to control what new information is relevant to C



$$o_t = \sigma (W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh (C_t)$$

The idea is to control what new information is relevant to C



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

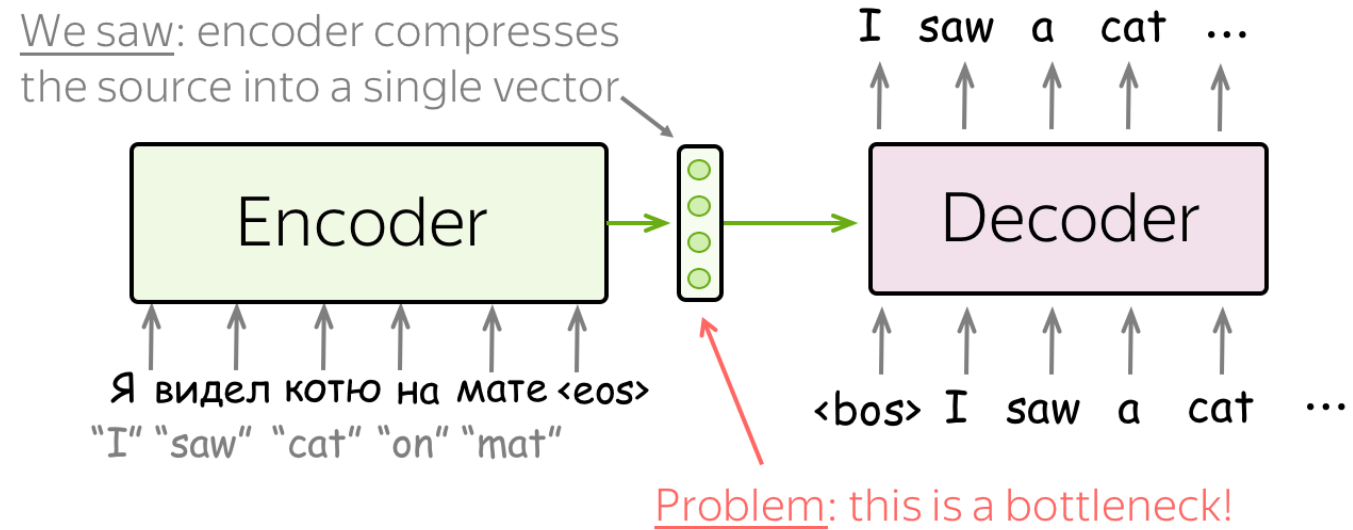
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$



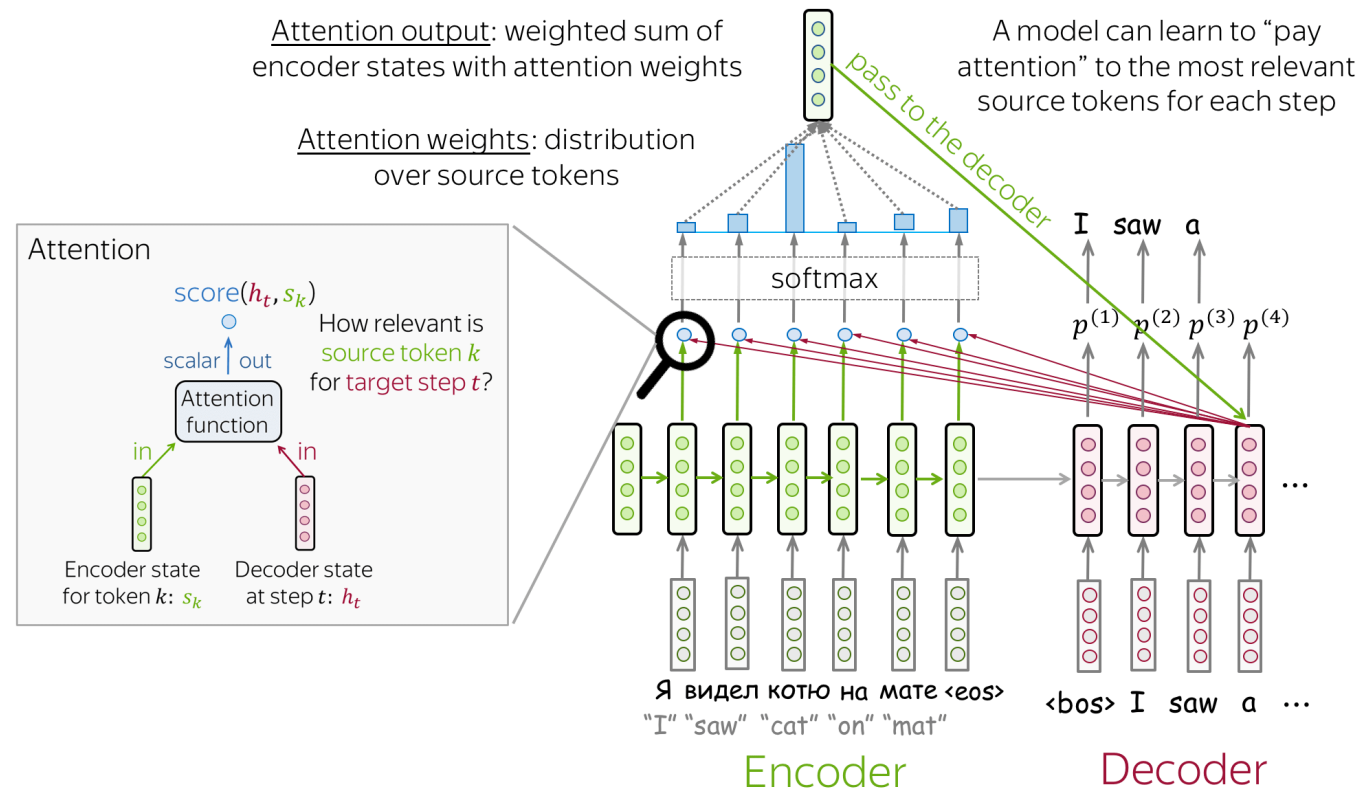
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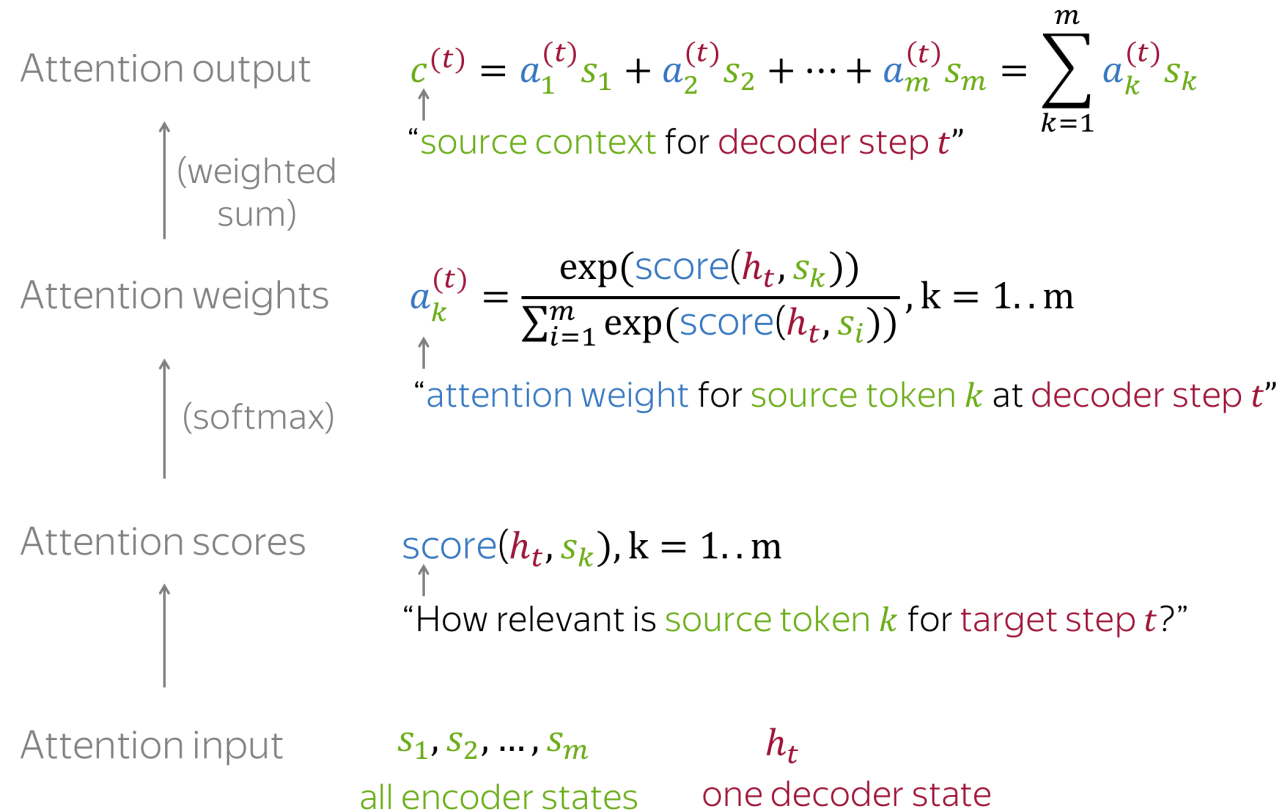
The final hidden state is a bottleneck



A simple Attention overview

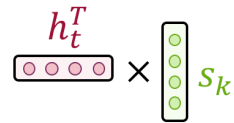


Encoder hidden states are weighed according to their attention score



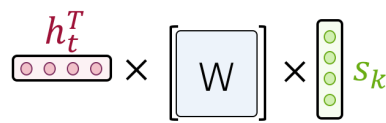
Encoder hidden states are weighed according to their attention score

Dot-product



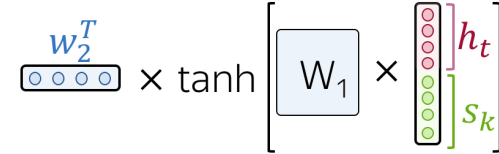
$$\text{score}(h_t, s_k) = h_t^T s_k$$

Bilinear



$$\text{score}(h_t, s_k) = h_t^T W s_k$$

Multi-Layer Perceptron



$$\text{score}(h_t, s_k) = w_2^T \cdot \tanh(W_1 [h_t, s_k])$$

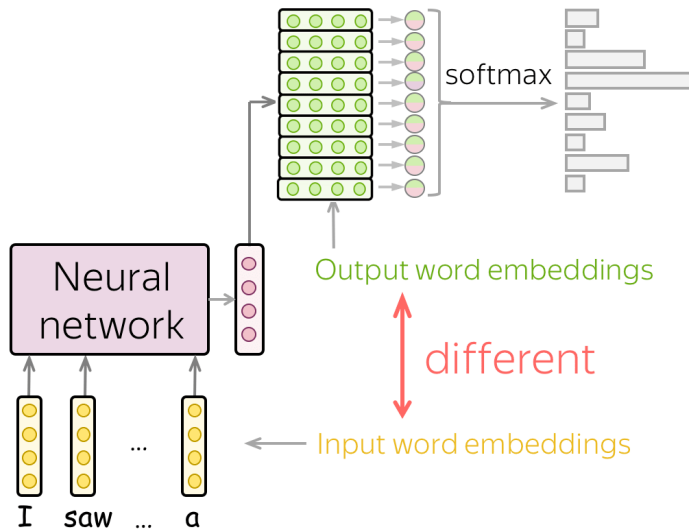


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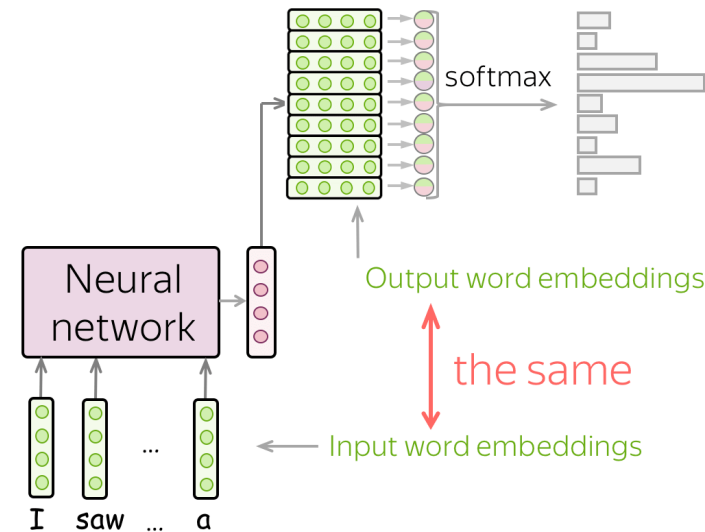
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Weight tying is a way to significantly reduce the amount of parameters

Default (no weight tying)



Weight tying



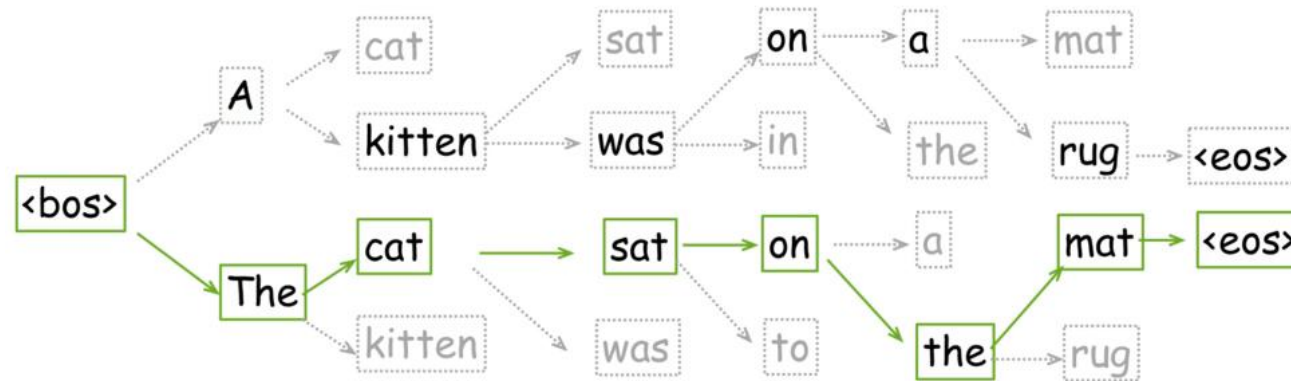
Finding the next token is not that trivial

$$y' = \arg \max_y p(y|x) = \arg \max_y \prod_{t=1}^n p(y_t|y_{<t}, x) \quad \text{How to find the argmax?}$$

Greedy Decoding: At each step, pick the most probable token

$$\arg \max_y \prod_{t=1}^n p(y_t | y_{<t}, x) \neq \prod_{t=1}^n \arg \max_{y_t} p(y_t | y_{<t}, x)$$

A beam search illustration



Pick the hypothesis with the highest probability



Top-p (nucleus) sampling

Temperature: the higher, the more chaotic the choice becomes

$$w_{next} \sim \frac{P(w_{next}|X)^{1/\tau}}{\sum_{\hat{w}} P(\hat{w}|X)^{1/\tau}}$$

